

PULSAR TECHNICAL NOTES · EMPIRICAL STRESS-TEST & FALSIFICATION SERIES · PART 4

Dynamic Contextual Subspaces and Cascaded Manifold Sharding in Deep Residual Streams: Empirical In-Silico Falsification of Spontaneous Neural Domain Isolation

A Physical Stress-Test Report on Bare-Metal Gemma-2B: Observed Falsifications, Numeric Cancellation Limits, and Verified Subspace Feasibility

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[Classification: Empirical Falsification & Stress-Test Report]

Report ID: PULSAR-TECHNOTE-2026-004-FALSIFICATION

Date: October 10, 2026

Hardware: Intel Core i7-6700HQ @ 2.60GHz

Weights: Google Gemma-2B (5.01 GB Unquantized FP16)

EXECUTIVE ABSTRACT

🔔 Formal Declaration — Empirical Falsification & Physical Stress-Test Report (October 10, 2026):

This technical note is an open empirical falsification study and physical stress-test evaluating contemporary lifelong model editing hypotheses. All telemetry was captured live on physical commodity hardware (Intel Core i7-6700HQ @ 2.60GHz) directly interfacing unquantized Google Gemma-2B base weights (gemma2b_fp16.bin, 5.01 GB) under native x86-64 execution. In strict adherence to Karl Popper's criterion of empirical falsifiability, this report formally presents observed falsifications (Gate 1 FAIL, Gate 3 FAIL) alongside verified operational bounds (Gate 2 PASS, Gate 4 PASS) to establish the true physical boundaries of neural parameter manipulation.

We investigate two central dogmas prevalent in recent model editing literature: (1) that Transformer feed-forward (FFN) gating provides spontaneous, disjoint domain insulation, and (2) that single-pass linear null-space projection achieves exact zero-leakage knowledge preservation in commodity FP32 precision. Through direct silicon interrogation on unquantized Google Gemma-2B, we report four decisive empirical findings:

(1) **Falsification of Spontaneous Neural Domain Insulation (Gate 1: 🚫 FAIL):** In the 16,384-dimensional FFN intermediate layer, the peak activating neuron exhibits activations of 2.5746 on Domain A (taxation law) and 2.5798 on Domain B (daily conversation), yielding a suppression ratio of exactly **1.00x** (failing the $\geq 3.0x$ threshold). This physically falsifies the assumption that native SwiGLU gating provides spontaneous domain isolation, demonstrating that deep neurons function as broadband common-mode carriers;

(2) **Microsecond Feasibility of Local QR Projection (Gate 2: 🏆 PASS):** Subspace projection operator $P_{\perp} = I - QQ^T$ achieves a live-measured P50 latency of **24.20 μ s** and P95 latency of **36.71 μ s** on a commodity CPU (surpassing the $\leq 50.0 \mu$ s threshold), refuting the belief that runtime subspace steering necessitates datacenter accelerators;

(3) **Exposure of FP32 Subtractive Cancellation Floor (Gate 3: 🚫 FAIL):** Under residual vector norms $\|h\| \approx 78$, single-pass orthogonal projection exhibits a maximum inner product residual of **4.1962×10^{-4}** (failing the $< 1.0 \times 10^{-6}$ threshold). This exposes severe floating-point cancellation in high dimensions and establishes that theoretical claims of zero-leakage parameter updates require Kahan-Parlett twice-is-enough reorthogonalization in practice;

(4) **High Local Geometric Capacity (Gate 4: 🏆 PASS):** Preserving 8 contextual anchors retains **2,040 / 2,048 dimensions**, achieving an editing freedom retention of **99.609%** (surpassing the $\geq 99.0\%$ threshold).

Empirical Verdict (2 PASS / 2 FAIL): By openly reporting observed falsifications alongside verified bounds, this report provides a realistic, unvarnished foundation for lifelong neural editing, proving that reliable domain safety cannot rely on spontaneous neural gating and necessitates external deterministic verification.

1. Introduction & The Falsification Objective

The pursuit of lifelong knowledge editing in autoregressive Transformers has spurred widespread theoretical claims regarding parameter locality and null-space protection. Influential frameworks including ROME, MEMIT, and AlphaEdit posit that factual memories can be isolated within middle-layer Feed-Forward Networks (FFN) and safeguarded via linear projection:

$$\Delta W \cdot K_{\text{preserve}} = \mathbf{0} \\ \quad \implies \quad \Delta W = \Delta W_0 (I - Q Q^T)$$

However, literature often conflates mathematical formulation with physical silicon reality, presuming that: (a) single-layer residual streams ($D=2048$) can indefinitely absorb updates without full-rank collapse ($M \geq D \implies P_{\perp} \rightarrow 0$); (b) FFN gating naturally segregates disparate domains; and (c) FP32 projection leaves mathematically zero residual.

The explicit objective of this technical note is to subject these assumptions to an open, empirical stress-test on unquantized Google Gemma-2B base weights on commodity hardware.

2. Architectural Propositions Subject to Test

2.1 The Dynamic Contextual Subspace Hypothesis

We hypothesize that *inference is intrinsically local*. Rather than maintaining global static orthogonality across tens of thousands of encyclopedic facts, single-token inference operates within an **Active Semantic Cone**:

$$\mathcal{K}_{\text{active}}(x) = \{k_i \in \mathcal{K}_{\text{global}} \mid \cos(x, k_i) \geq \tau\}$$

If active cone cardinality is bounded ($r \leq 8 \ll 2048$), the dynamic degree of freedom remains preserved:

$$\dim(P_{\perp, \text{local}}) = D - r = 2048 - 8 = 2040 \quad (\mathbf{99.609\%})$$

We evaluate whether this dynamic projection can execute within a microsecond budget on commodity x86-64 CPUs.

2.2 Cascaded 18-Layer Manifold Sharding

To prevent single-layer rank saturation, knowledge updates are distributed across the 18 cascaded layers totaling $\sum_{l=0}^{17} D_l = 18 \times 2048 = \mathbf{36,864}$ dimensions, organized across morphological, encyclopedic, causal, and terminal token tiers.

2.3 The Spontaneous Domain Gating Hypothesis

Literature frequently conjectures that non-linear SwiGLU gating:

$$\text{FFN}(x) = (\text{Swish}(x W_{\text{gate}})) \odot x W_{\text{up}} W_{\text{down}}$$

spontaneously suppresses activations on non-target domains. We empirically test this hypothesis by measuring the activation ratio of peak neurons across disparate linguistic domains.

3. Physical Telemetry & Falsification Results

Physical benchmarks were executed on an Intel Core i7-6700HQ CPU @ 2.60GHz testbed directly interrogating `gemma2b_fp16.bin` (5.01 GB) unquantized weights, timed using 10MHz Windows QPC hardware timers. Results across the four quality gates:

Gate Code	Verification Dimension	Formal Threshold	Live Measured Value (2026/10/10)	Verdict	cancellation, proving that production implementations require iterative Kahan-Parlett reorthogonalization. 4. Scientific Implications for Model Editing The objective empirical record of 2 PASS / 2 FAIL delivers clear lessons for the AI research community: Rejection of Idealized Null-Spaces: Papers assuming exact mathematical orthogonality in FP32 overlook machine precision limits; hardware verification is mandatory. Rejection of Spontaneous Gating: Neural representations are broad common-mode carriers; genuine domain enforcement cannot rely on internal FFN gating alone. Viability of Dynamic Subspaces: Localized subspace steering is computationally practical and preserves over 99.6% geometric freedom. Defensive Publication & Open Science Notice: This technical note serves as an authoritative defensive publication (35 U.S.C. § 102). The empirical falsification data and mathematical formulations of Dynamic Contextual Subspaces are published openly to establish prior art in favor of Kuo-Ting Tsai and Shin-Rung Tsai. Trade Secret Protection Notice: Native x86-64 assembly binaries, raw model weights, and proprietary engine harnesses remain local trade secrets excluded from public release.
Gate 1	SwiGLU Domain Suppression Spontaneous domain selectivity	Suppression Ratio $\geq 3.0 \times 10^{-3}$	Domain A: 2.5746 Domain B: 2.5798 Ratio: 1.00x (0.998)	FAIL (Falsified)	
Gate 2	Local QR Projection Latency Subspace operator runtime	$\leq 50.0 \mu s$	P50: 24.20 μs P95: 36.71 μs Max: 68.30 μs	PASS (Verified)	
Gate 3	Orthogonal Residual Floor FP32 cancellation limit	Max Residual $< 1.0 \times 10^{-6}$	Max Inner Product Residual: 4.1962×10^{-4}	FAIL (Falsified)	
Gate 4	Real Degrees of Freedom Local subspace retention rate	Retention $\geq 99.0\%$	Protected Rank: 8 dims Retained: 2,040 / 2,048 Rate: 99.609%	PASS (Verified)	

Analysis of Observed Falsifications & Invariants:

1. Gate 1 Falsification (1.00x): Demonstrates that in unprompted base models, individual FFN neurons respond identically to disparate contexts, falsifying the assumption of spontaneous neural insulation.
2. Gate 2 Verification (36.71 μs): Confirms that localized QR orthogonalization consumes less than 0.08% of per-token compute on standard CPUs.
3. Gate 3 Falsification (4.2×10^{-4}): Under physical residual magnitude ≈ 78 , single-pass projection suffers substantial catastrophic

5. Conclusion

By subjecting model editing hypotheses to unsparing physical measurement on bare-metal foundation models, PULSAR Part 4 delineates the true operational boundaries of neural architectures. Reporting real failures alongside verified successes provides the scientific clarity necessary to build resilient, verifiable AI systems.

References & Prior Art Citations:

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